An Exact Extension and an Infinite Family for OEIS A249106

Carlo Corti
Independent researcher
carlo.corti@outlook.com

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Abstract

OEIS A249106 lists integers whose squares, together with the squares of their first four arithmetic derivatives, sum to a square. An exact C enumeration of the inclusive interval $[434325392, 10^{10}]$, comprising 9565674609 integers, found precisely 1102826289 and 3945594513. Relative to the accepted five-term OEIS prefix, these are the sixth and seventh terms. Their positive witnesses have independent trial-division checks; the exclusion of all other inputs relies on the reviewed enumeration algorithm and its complete segment records, rather than a second independent scan. Separately, the fixed-point identity $\mathrm{D}(p^p) = p^p$ yields the family $4791p^p$ for every prime p. An explicit seed calculation and an induction prove membership and hence infinitude, without extrapolating the finite search. A general seed criterion also explains analogous constructions at other chain lengths, including $128p^p$ for A230543. The fixed-point machinery is historical; absolute priority for the seed specializations is not asserted. Combining the A249106 family with the enumeration bounds the eighth term by $10^{10} < a(8) \le 1366928213897301$; its exact value remains undetermined.

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1 Definition and context

The arithmetic derivative on nonnegative integers is determined by $\mathrm{D}(0) = \mathrm{D}(1) = 0$, $\mathrm{D}(p) = 1$ for primes p, and the product rule. For $x \ge 2$, unique factorization gives

$$x = \prod_p p^{e_p}, \qquad \mathrm{D}(x) = \sum_{p \mid x} e_p \frac{x}{p}. \tag{1}$$

These elementary properties and the fixed points p^p are discussed by Ufnarovski and Åhlander [5, Theorem 6] and in OEIS A003415 [2]. The map is not additive. Iterated derivatives below always mean composition. Put $x_j = \mathrm{D}^j(n)$ for $0 \le j \le 4$, with $x_0 = n$, and define

$$Q(n) = \sum_{j=0}^{4} x_j^2. \tag{2}$$

A249106 is the increasing enumeration, with index offset one, of the integers $n \ge 2$ for which $Q(n) = r^2$ for an integer $r \ge 0$ [1]. The restriction $n \ge 2$ follows the entry's enumeration convention: $n = 1$ would satisfy the bare equation trivially, since $Q(1) = 1$. Its Pythagorean sextuple is $(x_0, x_1, x_2, x_3, x_4, r)$: five chain coordinates and the root, with no primitivity requirement.

Paolo P. Lava introduced the sequence on October 21, 2014. The entry snapshot accepted for this project credits Ray Chandler with the fourth term on December 23, 2016 and the fifth on January 11, 2017. That snapshot has the five-term prefix

$$19164,\ 129357,\ 14971875,\ 45316123,\ 434325391. \tag{3}$$

The archived entry snapshot agrees with the public OEIS entry, whose published prefix is (3). The two additional terms reported here have not been submitted to OEIS. The prefix was checked at its five positive inputs; a full enumeration of $[2, 434325391]$ was not repeated.

Related entries concern different chain lengths or primitivity conditions: A210503 imposes a primitive three-coordinate condition; A230543 concerns four-coordinate tuples without that requirement. A249105, A249107 and A249110 concern five-, seven- and ten-coordinate Pythagorean tuples [4]. They provide context rather than inclusion relations with A249106. General polynomial parametrizations of Pythagorean sextuples [7] do not enforce the successive arithmetic-derivative constraints in (2).

The results here have two distinct sources. The finite search extends the indexed prefix through seven and supplies an exclusion bound. Subsequent analysis of four scaled witnesses gives an infinite family by proof. The fixed-point identity used in that proof is prior arithmetic-derivative theory; no absolute priority claim is made for its specialization to this seed. The project artifacts are publicly available from the GitHub repository and Zenodo archive cited in reference [11]. The designated Zenodo DOI identifies the release series rather than one immutable deposited version.

2 Finite enumeration and witnesses

The search began immediately after the accepted fifth term and stopped at 10^{10}. Seven completed segments partition the interval, as shown in Table 1. The two phases used blocks of 2^{20} inputs and eight threads. Phase 1 consisted of one segment configured for at most 540 blocks and clipped at 10^9. Phase 2 used segments of 1500 blocks, with the last clipped at 10^{10}.

Phase/segment	First input	Last input	Hits
1/0	434325392	1000000000	0
2/0	1000000001	2572864000	1
2/1	2572864001	4145728000	1
2/2	4145728001	5718592000	0
2/3	5718592001	7291456000	0
2/4	7291456001	8864320000	0
2/5	8864320001	10000000000	0

Table 1: Completed inclusive segments of the finite search. The hit counts come from the enumeration records; the segment checks are not independent re-enumerations of non-hits.

The interval cardinality is exactly

$$10^{10} - 434325392 + 1 = 9565674609. \tag{4}$$

Table 2 gives the entire consolidated prefix and all witness coordinates. Its first five rows derive from previously validated outputs of an earlier C build; the mathematical source regions are byte-identical in the final campaign build. The last two rows are campaign outputs that also agree with independently computed Python witnesses.

Theorem 1 (Finite computational result)**.** *Subject to the accepted prefix* (3) *and the correctness and complete execution of the enumeration described in Section 5, the sixth and seventh terms are*

$$a(6) = 1102826289, \qquad a(7) = 3945594513,$$

and there is no further member at most 10^{10}.

i	$a(i)$	$\mathrm{D}(a(i))$	$\mathrm{D}^2(a(i))$	$\mathrm{D}^3(a(i))$	$\mathrm{D}^4(a(i))$	r
1	19164	25564	31848	58412	61916	96336
2	129357	172557	214974	394281	417933	650268
3	14971875	19971875	24881250	45634375	48371875	75262500
4	45316123	198116	198120	419644	419648	45320875
5	434325391	3315592	5447100	9543040	36609088	436016633
6	1102826289	411955468	411955472	830107536	1973010320	2477395545
7	3945594513	5263263313	6557049366	12026198429	12747622097	19834209612

Table 2: Seven indexed witnesses. In every row the sum of squares of the five chain coordinates equals r^2.

Computational evidence. The seven completed records have adjacent endpoints and total cardinality (4). The program audits adjacency within each phase separately; the join from 10^9 to $10^9 + 1$ is checked externally against the two configurations and the completed records. Their combined hit list consists exactly of the two values displayed. Both positive chains were separately recomputed by exhaustive trial division and integer square-root tests. All remaining inputs are excluded by the C enumeration, whose coverage and arithmetic invariants are given in Section 5. File-integrity and native verification checks establish the consistency of the completed records, but do not constitute an independent second enumeration. Combining this finite result with the accepted earlier prefix gives the asserted indices. □

3 An unconditional infinite family for A249106

3.1 From observations to a statement for every prime

The first three terms and the seventh have the respective multipliers $2^2, 3^3, 5^5, 7^7$ of $4791 = 3 \cdot 1597$. This observation admits an exact extension to every prime, using the known identity $\mathrm{D}(p^p) = p^p$. The observations suggested the seed; they do not prove the universal statement. The proof uses only the defining product rule, four small factorizations and one integer square identity. It applies to every prime, including primes beyond any computational search limit.

3.2 The historical fixed-point mechanism

For a prime p and integer $k \geq 1$, the product rule gives $\mathrm{D}(p^k) = kp^{k-1}$, hence $\mathrm{D}(p^p) = p^p$. These fixed points are established prior theory [5, Theorem 6]. The transition $T(m) = m + \mathrm{D}(m)$ is itself OEIS A129283 [3]. Its connection with multiplication by p^p was recorded in A003415 by R. Zumkeller on April 7, 2007 [2]. Its extension to a whole derivative chain follows by induction.

Lemma 2. *Let t be a positive integer with $\mathrm{D}(t) = t$, and put $T(m) = m + \mathrm{D}(m)$ on nonnegative integers. For every $m \geq 0$ and $j \geq 0$,*

$$\mathrm{D}^j(tm) = tT^j(m). \tag{5}$$

Proof. The case $j = 0$ is immediate. If it holds at j, the product rule gives

$$\mathrm{D}(tT^j(m)) = \mathrm{D}(t)T^j(m) + t\,\mathrm{D}(T^j(m)) = tT^{j+1}(m).$$

This proves the induction. No coprimality condition is needed. □

For $t = p^p$ and $m > 1$, the one-step identity appears in Ufnarovski and Åhlander [5, Theorem 4], whose divergence statement already uses iteration of that identity. By their Theorem 6, the positive fixed points t in Lemma 2 are exactly p^p. The lemma states the iterate formula explicitly and includes $m = 0, 1$. Since D is not additive, T^j is a composition, not a binomial expansion in powers of D. For example, $T^2(m) = m + \mathrm{D}(m) + \mathrm{D}(m + \mathrm{D}(m))$; replacing the last derivative by a sum of derivatives would be unjustified. Shared factors are allowed: $p = 3$ divides the seed, and other primes can divide later seed coordinates.

3.3 The seed and the unconditional family

Theorem 3 (Unconditional family)**.** *For every prime p, the integer $4791p^p$ belongs to A249106, with witness*

$$p^p(4791, 6391, 7962, 14603, 15479, 24084). \tag{6}$$

In particular, A249106 is infinite.

Proof. The following four factorizations give the seed calculation directly:

$$\begin{aligned} 4791 &= 3 \cdot 1597, & \mathrm{D}(4791) &= 1597 + 3 = 1600, \\ 6391 &= 7 \cdot 11 \cdot 83, & \mathrm{D}(6391) &= 913 + 581 + 77 = 1571, \\ 7962 &= 2 \cdot 3 \cdot 1327, & \mathrm{D}(7962) &= 3981 + 2654 + 6 = 6641, \\ 14603 &= 17 \cdot 859, & \mathrm{D}(14603) &= 859 + 17 = 876. \end{aligned}$$

All displayed factors are prime, as can be checked by trial division through their integer square roots. Hence successive applications of T yield $(4791, 6391, 7962, 14603, 15479)$, and

$$4791^2 + 6391^2 + 7962^2 + 14603^2 + 15479^2 = 580039056 = 24084^2. \tag{7}$$

For prime p, equation (1) gives $\mathrm{D}(p^p) = p^p$. Lemma 2 and (7) prove (6). Explicitly, the four derivatives of $4791t$, where $t = p^p$, are $6391t$, $7962t$, $14603t$ and $15479t$, so

$$Q(4791p^p) = p^{2p}\, 580039056 = (24084p^p)^2.$$

This is exactly the defining condition of A249106. The integers $4791p^p$ are distinct and unbounded as p runs over the infinitely many primes, proving infinitude. □

If $q > p \geq 2$, then $q^q > p^q > p^p$, so increasing prime parameters give increasing family members. Theorem 3 has no hypothesis about the accepted prefix, completion of the C campaign, a primality conjecture, machine-word sizes or availability of a public software deposit. Its validity follows from the stated definition of D. The indexed finite result in Theorem 1 has different dependencies.

3.4 Primitivity, scope and unknown indices

The seed sextuple is primitive: already $\gcd(4791, 6391) = 1$. Thus the common divisor of all six coordinates in (6) is exactly p^p. For rows 1 through 7 of Table 2, the six-coordinate greatest common divisors are respectively

$$4,\ 27,\ 3125,\ 1,\ 1,\ 1,\ 823543.$$

In particular, rows 4, 5 and 6 are primitive witnesses outside this scaled family. Their first coordinates are not divisible by 1597. Neither that divisibility nor the family is an exhaustive search filter. The primes $2, 3, 5, 7$ yield indices $1, 2, 3, 7$, so the prime parameter is not a formula for the sequence index.

The product-rule proof also shows why arbitrary multipliers cannot be substituted for p^p: in general $\mathrm{D}(sm) = s\,\mathrm{D}(m) + m\,\mathrm{D}(s)$, so the seed transition is T only when $\mathrm{D}(s) = s$. The theorem supplies a sufficient family, not a classification of every multiplier or every member. The ratios $\mathrm{D}^j(4791p^p)/(4791p^p)$, $0 \leq j \leq 4$, are independent of p, and the root/input ratio is $24084/4791$. The entire witness changes by a common scale, while its normalized shape stays fixed.

Table 3 gives four further consequences. Here n_p is indexed by its prime parameter, not $a(p)$ and not an assigned OEIS index. Each root is $r_p = 24084p^p$ and the chain is given by (6); the intervening intervals have not been enumerated.

p	$n_p = 4791p^p$
11	1366928213897301
13	1451074635683484123
17	396330809469743943717 2007
19	947860857026856240549298 3389

Table 3: Four proven members without assigned sequence indices. These symbolic consequences do not extend the indexed b-file.

3.5 A bound without identification of the next term

Corollary 4. *The upper bound $a(8) \le 4791 \cdot 11^{11}$ holds independently of the finite exclusion and completeness of the accepted baseline. With the finite-result assumptions of Theorem 1, the combined bounds are*

$$10^{10} < a(8) \le 4791 \cdot 11^{11} = 1366928213897301. \tag{8}$$

Proof. The seven distinct positive members in Table 2, whose chains were recomputed by Python trial division as described in Section 6, and the $p = 11$ member of Theorem 3 give eight distinct members at most $4791 \cdot 11^{11}$. This proves the upper bound without assuming that the table contains every earlier member. The $p = 11$ member exceeds 10^{10} and has root $24084 \cdot 11^{11} = 6871446274995324$. The lower bound follows from Theorem 1. □

This upper-bound witness is derived symbolically, has no assigned sequence index, and was not found by extending the computational search. It even exceeds the implementation's conservative input cap discussed below.

4 Related families and the scope of the contribution

4.1 A criterion for any derivative-chain length

The same proof works for other chain lengths, but each requires its own seed identity. For integers $h \ge 1$ and $m \ge 1$, define

$$Q_h(n) = \sum_{j=0}^{h} \left(\mathrm{D}^j(n)\right)^2, \qquad S_h(m) = \sum_{j=0}^{h} \left(T^j(m)\right)^2.$$

Lemma 2 gives, for every prime p,

$$Q_h(mp^p) = p^{2p} S_h(m). \tag{9}$$

Thus mp^p has a square sum at depth h if and only if $S_h(m)$ is an integer square. For the converse, if $r^2 = t^2 S_h(m)$ with $t = p^p$, comparison of prime exponents shows $t \mid r$, so $S_h(m) = (r/t)^2$. The forward implication follows by multiplying an integer seed root by t. One exact seed calculation therefore proves membership for every prime; failure rules out all prime parameters for that seed and depth. This elementary criterion is derived here from the known product rule, without a claim that the scaling principle is new. Here $Q = Q_4$, and for A249106 the successful test is $S_4(4791) = 24084^2$.

The seed also has a conditional minimality property. If $m \ge 1$ and $S_4(m)$ is a square, the case $p = 2$ gives the member $4m$ of A249106. Thus, assuming the accepted first term $a(1) = 19164$ is minimal, $4m \ge 19164$ and $m \ge 4791$. Since 4791 passes the seed test, it is then the least depth-four seed. Similarly, assuming the recorded first member 512 of A230543 is minimal, its least depth-two seed is 128: every such seed yields a member $4m \ge 512$, and 128 passes the test in Section 4.3. These are deductions from the respective first-term assumptions, not new independent exclusions of smaller inputs.

4.2 Established analogues in related OEIS entries

For $m = 1$, all iterates of T equal one and $S_h(1) = h + 1$. Whenever $h + 1 = b^2$ for an integer b, every p^p has a constant derivative chain and root bp^p. In particular,

$$Q_3(p^p) = (2p^p)^2, \qquad Q_8(p^p) = (3p^p)^2.$$

The inclusions of A051674, the sequence p^p for prime p, are already recorded in A249105 and A249110 [4, 3]. They prove infinitude of those sequences without a prime-pair conjecture. At depth four the seed $m = 1$ fails because $S_4(1) = 5$ is not a square; the nonconstant seed 4791 supplies the needed replacement.

A210503 has a primitive condition. Its entry records the twin-prime construction [4]: when p and $p + 2$ are primes,

$$n = p(p + 2), \qquad \mathrm{D}(n) = 2(p + 1), \qquad n^2 + \mathrm{D}(n)^2 = (n + 2)^2.$$

Here p and n are odd, and neither p nor $p+2$ divides $2(p+1)$, so $\gcd(n, \mathrm{D}(n)) = 1$ and the triple is primitive. The identity follows from $n = (p + 1)^2 - 1$. Each prime pair gives a member, but infinitude from this construction would require infinitely many twin primes. The scaled witnesses of Theorem 3 have common divisor p^p and cannot provide primitive triples simply by dropping coordinates.

4.3 An explicit comparison for A230543

A230543 uses depth two with no primitivity restriction [4]. Since $128 = 2^7$ and $576 = 2^6\,3^2$,

$$\mathrm{D}(128) = 448, \quad \mathrm{D}(576) = 2112, \quad (T^0(128), T(128), T^2(128)) = (128, 576, 2688).$$

The square identity

$$128^2 + 576^2 + 2688^2 = 7573504 = 2752^2$$

and (9) prove that every prime p gives the A230543 witness $p^p(128, 576, 2688, 2752)$. The cases $p = 2, 3, 5, 7$ give $512, 3456, 400000, 105413504$, all present in the consulted entry. This family is derived in the comparison: the consulted entry does not state the generating formula, and no priority claim is made for it. Its membership and infinitude follow from the proof independently of the recorded examples.

For A249107 (depth five), the consulted entry supplies finite examples but no analogous generating family. This describes that entry, not the absence of a theorem elsewhere. Different depths give no inclusion into A249106 merely by deleting or appending derivatives.

4.4 Prior art and what is established here

The historical ingredients are the fixed points p^p, the identity for $\mathrm{D}(mp^p)$ and the transition $m+\mathrm{D}(m)$ [5, 2, 3]. Kovič also discusses fixed points and higher derivatives in a broader differential-equation setting [6, Sections 1.2 and 2.1]. The contribution made explicit in Theorem 3 is the specialization to the verified seed 4791, its complete chain witness and the resulting unconditional infinitude of A249106. It was proved after the finite campaign, rather than used as a filter.

A targeted primary-source and web search on October 3, 2026 did not identify an earlier explicit statement of the family $4791p^p$ or its displayed seed witness. This limited search does not establish absolute priority: indexing gaps and literature unavailable in full text can hide precedents. In particular, the 2013 Hoepli book by G. Balzarotti and P. P. Lava, *La derivata aritmetica*, listed in A003415 [2], was not available to this project in full text and was not consulted. The family is therefore presented with its proof and known antecedents, without asserting a first discovery of the underlying mechanism or excluding an earlier occurrence of this particular family.

5 Exact enumeration algorithm

The campaign used a C17 program with an unsigned 128-bit integer extension, GMP integer arithmetic, OpenMP, and OpenSSL digest routines. The final code does not use FLINT. Its mathematical stages are segmented evaluation of the first derivative, exact factorization of the next three operands, and an exact square test. Operational segments group consecutive blocks; the mathematical processing within each block is independent of this grouping.

5.1 Consecutive inputs and the first derivative

For each operational segment the program precomputes the primes through the integer square root of the segment upper endpoint. Each block processes only the table prefix satisfying $p^2 \le H$, where H is the block upper endpoint. Within a block, a residual array starts at the original inputs n, and a derivative accumulator starts at zero. For each processed prime p, all block multiples of p are visited; the full multiplicity e is removed from their residuals, and $e(n/p)$ is added to their accumulators. At every stage, the product of the residual and the removed prime powers is n, and the accumulator is precisely the contribution of those removed powers in (1).

After all primes through the block square root have been processed, any residual greater than one is prime: a composite residual would have a prime factor no larger than its square root, hence no larger than that of n, and would already have been divided out. Its remaining contribution n/q is then added. This proves exact evaluation of $\mathrm{D}(n)$ for every block input.

The implemented early exclusion is the prime-input case. For $n \ge 2$, $\mathrm{D}(n) = 1$ holds exactly when n is prime. Composite inputs have at least two prime factors counted with multiplicity and derivative greater than one. For a prime, the chain is $(n, 1, 0, 0, 0)$, and $n^2 < n^2 + 1 < (n+1)^2$. Such inputs cannot qualify. The code uses no filter requiring the factor 1597 and no separately implemented congruence filter.

5.2 Derived operands and deterministic completion

Each of x_1, x_2, x_3 is factored to obtain the next derivative. The word-size primality routine applies small-prime divisions followed by the strong probable-prime test with bases

$$2,\ 325,\ 9375,\ 28178,\ 450775,\ 9780504,\ 1795265022.$$

These seven bases are sufficient for deterministic primality below 2^{64}, as recorded by Forišek and Jančina [8, Theorem 3]. Each base is reduced modulo the operand, with zero residues skipped. Modular products use unsigned 128-bit arithmetic, so they are exact for word-size operands.

The splitter first checks small divisors through 97. It then tries 32 Pollard-rho-style trajectories [9], with the recurrence $z \mapsto z^2 + c \pmod{n}$, Floyd cycle comparison and greatest common divisors, using at most 100000 steps per trajectory. This is an acceleration rather than a correctness assumption. If the trajectories do not split a known composite, exhaustive odd trial division from 101 through its square root supplies a deterministic fallback. The condition $d \le n/d$ avoids overflow in the loop bound.

A proper factor splits the problem recursively into two smaller positive operands. Every accepted leaf is checked prime, all multiplicities are collected, and the reconstructed product is required to equal the original operand. There are at most 63 prime factors counted with multiplicity for a positive operand below 2^{64}, so the 64-entry factor array suffices. Reconstruction multiplies only a partial product by one word-size factor at a time in the wider type. Equation (1) then gives the derivative exactly. Operands zero and one produce zero directly. Termination follows from strictly smaller recursive operands and finite fallback trial division. An interrupt or unsuccessful arithmetic operation leaves a block incomplete; it is never recorded as a negative result.

5.3 Integer bounds and the square test

For $x \geq 2$, if $\Omega(x) = \sum e_p$, then

$$\mathrm{D}(x) \leq \frac{x\Omega(x)}{2} \leq \frac{x\lfloor \log_2 x \rfloor}{2}. \tag{10}$$

The first inequality follows from $p \geq 2$ and the second from $2^{\Omega(x)} \leq x$. This is the elementary derivative bound underlying the width analysis; its logarithmic version appears in [5, Theorem 9]. The integer envelope $F(U) = \lfloor U\lfloor \log_2 U \rfloor/2 \rfloor$ is nondecreasing for $U \geq 2$. Starting from $U_0 = 9 \cdot 10^{14}$ and applying $U_{j+1} = F(U_j)$ gives

$$\begin{aligned} U_1 &= 22050000000000000, \\ U_2 &= 595350000000000000, \\ U_3 &= 17562825000000000000 < 2^{64}, \\ U_4 &= 553228987500000000000 < 2^{128}. \end{aligned}$$

The zero and one cases also satisfy these envelopes. Thus x_0 through x_3 fit the word type, while x_4 uses the wider type. Even the conservative bound $\sum U_j^2$ needs 138 bits: the square sum is therefore formed in GMP, not in a fixed 128-bit accumulator. GMP's integer perfect-square predicate and square root [10] decide (2); the root is checked on export to the wider type. There is no floating-point membership decision. Indeed $r \leq \sqrt{5}\, U_4 < 2^{71}$ in this envelope.

Additional program constraints support this domain on the measured 64-bit platform. Blocks have a power-of-two width between 2^{14} and 2^{20}, thread counts are between one and eight, and segment counts are below 2^{40}. The prime table bound is at most 30000000, fitting its 32-bit entries. A block uses two 64-bit arrays and one byte array, at most 17 MiB altogether, in addition to the prime table and worker state. Segment-width and endpoint calculations use the wider type: a word-size block-group count times a 2^{20} block is below 2^{84}, and multiplying by a segment identifier below 2^{40} remains below 2^{124}, with the bounded endpoint additions also safe. Allocation, I/O and elapsed-time errors cause failure rather than certification. The cap is a sufficient numerical domain, not a maximum mathematical domain, a runtime estimate, or a claim of feasible enumeration to that endpoint. For comparison, the same sufficient envelope at 10^{15} already gives $U_3 = 19514250000000000000 > 2^{64}$.

5.4 Parallel blocks and persistent coverage

The residual sieve is followed by an OpenMP loop with dynamic scheduling. Each input owns a separate result flag, and worker-local factorization and GMP state avoid shared arithmetic mutation. A lock-free atomic stop flag communicates interruption. All inputs must reach a complete positive or negative state before the block is accepted. Positive flags are reevaluated serially and emitted in increasing input order. A failed or interrupted block does not advance the saved frontier.

Checkpoints persist only fully processed blocks. A segment is sealed only when its frontier reaches its specified upper endpoint. Each saved state contains its bounds, frontier and positive rows, together with a digest of its serialized content; a separate seal binds the complete segment file. Saved configuration binds the source, build recipe and executable identities. Exclusive advisory locking prevents simultaneous cooperating writers, while readers use shared locks. Atomic replacement uses an exclusive temporary file, flushing, file synchronization, rename, and directory synchronization; initialization also synchronizes the parent directory.

On restart, the program audits the stored records and resumes after the last saved complete block. Work in an unsaved partial block is repeated. This preserves coverage without promising resumability inside an input's factorization. If a seal was committed before its aggregate tables, a valid short aggregate prefix is classified as repairable and rebuilt from the segment authority; a

changed row is rejected. Within each run, sealed segment endpoints and their adjacency establish coverage rather than an aggregate count alone. The two campaign phases are separate runs; their cross-phase join and combined cardinality are checked externally from the configurations and completed records. These guarantees assume a cooperating local-filesystem environment. They do not establish resistance to forged unkeyed digests, arbitrary hardware faults, noncooperating writers, or every power-loss/filesystem behavior.

The native verification command audits identities, framing, bounds, seals, ordering and aggregate consistency, and reevaluates the stored positive witnesses with the same arithmetic backend. It does not process every excluded input again. Accordingly, it corroborates persistent coverage and positive consistency but is correlated with the enumeration kernel.

5.5 Cost and reproducibility

Sieving consecutive original inputs reduces repeated factorization of those inputs. The derivative values themselves are not a consecutive interval. The cost includes prime-table construction per segment, block sieving, factorization of derived operands, exact square tests, and persistent-state audits. In particular, deterministic trial division can dominate difficult fallback cases. No global $O(N \log\log N)$ claim is made for the complete predicate. Aggregate rebuilding also grows with the number of sealed records.

The source and build recipe specify C17 with optimization level three, strict warnings, OpenMP, and linkage to GMP and OpenSSL. Independent positive validators use Python integer arithmetic and exhaustive trial division, followed by an integer square root and an exact equality test. The earlier sixth-term validator must read the frozen one-hit state supplied to it; the later two-hit state is a different protocol input. The seventh-term validator imports that same trial-division routine, requires textual identity of the recorded first hit with the frozen, previously validated sixth-term input, and recomputes only the new seventh-term hit. These two validators are independent of the C factorization backend, but share their own Python arithmetic routine and the defining equation. The public deposit includes the frozen inputs as well as the scripts; a script alone cannot replay the historical protocol states.

6 Validation and measured execution

The independent positive checks factored four operands for each new term, computed the four derivatives, and tested the resulting sum with an exact integer root. Each historical check requested and produced one distinct new positive witness. For the sixth term the operand factorizations are

$$x_0 = 3 \cdot 43 \cdot 59 \cdot 144899, \qquad x_1 = 2^2 \cdot 102988867,$$
$$x_2 = 2^4 \cdot 109 \cdot 199 \cdot 1187, \qquad x_3 = 2^4 \cdot 3 \cdot 23 \cdot 751909.$$

For the seventh they are

$$x_0 = 3 \cdot 7^7 \cdot 1597, \qquad x_1 = 7^8 \cdot 11 \cdot 83,$$
$$x_2 = 2 \cdot 3 \cdot 7^7 \cdot 1327, \qquad x_3 = 7^7 \cdot 17 \cdot 859.$$

The square sums are respectively 6137488686385847025 and 393395870932753190544. These checks verify membership, not the absence of smaller members.

Implementation checks retained in the project include the five baseline singletons, each compared with the SymPy oracle, and independent SymPy-based hit-set comparisons over $[2, 500]$, $[19150, 19180]$ and $[129340, 129370]$, respectively 499, 31 and 31 inputs. One-, four- and eight-thread comparisons on a low window and a 16384-input window ending at the technical cap tested deterministic output; they are comparisons of the same program, not independent oracles

for the whole cap domain. Segment-grouping and interrupted-resume tests checked the coverage contract, while altered rows, digests and frontiers were required to fail. A forced-fallback harness exercised the factorization fallback and wide-root paths.

These arithmetic and broad operational tests ran on the earlier build. Their transfer to the campaign build is restricted to the source regions that are byte-identical. The final change added an initialization path-length guard; it did not change arithmetic, block processing or thread behavior. Targeted final-build tests exercised that guard and adjacent I/O error paths. AddressSanitizer and UndefinedBehaviorSanitizer checks on the earlier build covered the first singleton and the small initial interval, with leak sanitization unavailable. A separate Clang thread-sanitizer check exercised a four-thread small interval with OpenMP instrumentation and a race-detection control; it was not a whole-campaign or final-build thread-sanitizer run. Final-build initialization sanitizer checks supplied additional bounded evidence. These checks detect particular implementation faults; none replaces the domain proof or independently enumerates billions of negative cases.

During manuscript preparation, a bounded seed check verified the four small factorizations in Theorem 3 and their agreement with existing rows 1, 2, 3 and 7. A bounded Python trial-division check also recomputed all seven chains and their roots, independently of the C factorization backend; it used the same Python factorization routine as the historical positive validators. Another finite integer check verified the seven table square identities, segment adjacency and counts, timing sum and width envelopes. These are mathematical and bookkeeping checks, not extensions of the official search. The expanded exposition includes a bounded check of the eight prime parameters $2, 3, 5, 7, 11, 13, 17, 19$, the depth-two seed 128, constant fixed-point chains and three twin-prime examples. Direct trial-division derivatives confirm the displayed positive chains. These checks corroborate examples; the general statements depend on their proofs, and no new sequence indices are certified.

The author executed the campaign on an AMD Ryzen 9 7940HS processor running Linux Mint 22.3, with Linux kernel 6.17.0-42-generic, x86-64, glibc 2.39 and GCC 13.3.0. The recorded runtime libraries include GMP, OpenSSL's crypto library and GNU OpenMP. All seven externally timed segment-advance calls exited zero. Their elapsed times sum to approximately 7984 seconds, or 2 hours 13 minutes 4 seconds, using eight threads. This total includes the work performed inside those calls and excludes unrecorded initialization, separate audits and manual activity. Dividing (4) by the recorded elapsed time gives approximately $1.20 \cdot 10^6$ original inputs per second for this campaign. This is an aggregate measurement, not a constant per-input cost or an extrapolation to another decade or to the technical cap.

7 Consequences and limits

Relative to the accepted five-term baseline, the finite computation adds two consecutive terms and excludes additional members through 10^{10}. Its negative evidence rests on complete execution of the exact C algorithm; only positives and small windows have independent arithmetic comparisons. The full earlier prefix interval was not rescanned.

The independent mathematical consequence is stronger than finite density evidence: the explicit fixed-point family proves infinitude. It explains four witnesses in the consolidated prefix while leaving three primitive witnesses outside the family. Combining the proof and finite exclusion gives (8), but neither determines $a(8)$ nor rules out further members between 10^{10} and the displayed upper bound. No conjecture about density, successive indices or a complete classification is proposed. The criterion in Section 4 explains how the historical mechanism yields families at other depths. The examples and source comparison distinguish the mechanism from its particular specializations; the literature search does not certify absolute priority.

8 Data availability

The public GitHub and Zenodo package [11] contains the source code, build recipe, accepted baseline, indexed results, complete segment, environment and timing records, independent positive-check scripts with frozen inputs, bounded seed checks and validation summaries. The manuscript is supplied as monolithic LaTeX source and a matching PDF; its witness table is embedded here. Both providers' initial release payloads were downloaded, checked against the full inventory and independently passed the bounded validation command. This does not independently rescan negative intervals. The designated DOI identifies the release series; exact deposits have separate DOIs. Primary-source locators are included; third-party full texts, private research notes and external reviews are excluded.

9 AI use disclosure

Large language models were used as research and drafting assistants during the A249106 workflow, including OpenAI Codex and external LLM-based review systems from the ChatGPT, Claude, DeepSeek, Gemini, Grok, Qwen, MiniMax, Meta and Z.ai families. They are named at family level where exact model-version metadata were not consistently available in the local artifacts.

Mathematical analysis. OpenAI Codex assisted in reconstructing the definition, checking primary sources, analyzing the seed scaling identity, and organizing the proof and finite evidence. AI-generated preliminary triage reports and a stored NotebookLM synthesis informed earlier selection and strategy research; they were not treated as proofs or as substitutes for primary sources. One preliminary triage had noted the shared factor 1597 in the first three terms as an unproved pattern; the fixed-point family and its proof followed the finite campaign.

Algorithm and code. OpenAI Codex assisted with implementation and local validation. The retained code-review reports include OpenAI, Anthropic Claude, Z.ai GLM-5.3, Google Gemini, xAI Grok, DeepSeek, Meta, MiniMax and Alibaba Qwen tools. Their roles were code and artifact review, with execution claims distinguished from static reasoning. Model versions are not inferred from filenames or behavior; the attribution of GLM-5.3 is to Z.ai. These reports concern code and do not constitute external review of this manuscript.

Manuscript. OpenAI Codex assisted in drafting the English text and in sequential internal mathematical, numerical, source and exposition checks. These passes share the drafting context and are not independent external reviews. The first full external manuscript round supplied nine reports from the ChatGPT, Claude, DeepSeek, Gemini, GLM, Grok, Meta, MiniMax and Qwen families. Their observations were checked against the source artifacts before incorporation; reported remote executions were not treated as locally reproduced evidence. A focused second round supplied reports from Claude and ChatGPT, leading to final clarifications of positive-validation provenance and citation wording. Family overlap with code review, and the OpenAI drafting/review relationship, limit independence. The author subsequently requested a fuller account of Theorem 3. OpenAI Codex assisted with the primary-source comparison, the general seed criterion and the A230543 example, followed by bounded arithmetic checks and sequential internal review. A focused third review of the expanded text, from the Claude family, found no scientific issue and suggested clarifications of attribution, scope and exposition. These were checked against the retained sources before incorporation; family overlap with earlier rounds limits independence. The final claims are based on source artifacts, recorded computations, independent positive validators and primary literature, not conversational assertions. Carlo Corti retains full scientific responsibility for the mathematical statements, implementation, numerical results, validation boundaries and conclusions.

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